

Rahmi Ramadhani Umar

Associate Professor of Mathematics Education, Universitas Potensi Utama, Medan
JESH Postdoctoral Fellow (Austrian Academy of Sciences), Johannes Kepler University Linz
Dr. in Mathematics Education (Universitas Negeri Medan, 2024)

rahmiramadhani3@gmail.com rahmiumar.github.io github.com/rahmiumar ORCID 0000-0002-8049-7234
Altenberger Straße 69, 4040 Linz, Austria Universitas Potensi Utama, Medan, Indonesia

27	9	119	31	6
Documents, Scopus	h-index, Scopus	Documents, Scholar	h-index, Scholar	Patents and IP

Profiles: Scopus 57202252421 · Google Scholar · Web of Science N-8427-2018 · SINTA 95483 (overall score 2252)

RESEARCH PROFILE

My research brings together technology-enhanced mathematics learning and educational measurement. I design and validate learning trajectories that pair GeoGebra applets with AI assistance, embed local culture into statistics and numeracy through the Ethno-Flipped Classroom model, and use Rasch Rating Scale and Many-Facet Rasch models to build and check the instruments that measure this learning.

Mathematics education · GeoGebra and AI-supported learning · Rasch measurement · Ethnomathematics · Numeracy · Teacher education

APPOINTMENTS

Jun 2026 – Nov 2026	JESH Postdoctoral Fellow / Visiting Scholar Johannes Kepler University Linz, Austria - Postdoctoral research on technology-enhanced mathematics education, focused on GeoGebra and AI integration. - Design and validation of learning trajectories that support numeracy development. - Joint projects and publications with international researchers; seminars and grant proposals.
Jun 2025 – present	Associate Professor Universitas Potensi Utama, Medan, Indonesia - Undergraduate and postgraduate courses in mathematics, statistics, and research methodology. - Research projects in ethnomathematics, numeracy, educational technology, and Rasch measurement. - Supervision of research projects, theses, and publications; competitive grants and collaborations. - Curriculum development, quality assurance, accreditation, and teacher professional development.
2019 – 2025	Senior Lecturer / Assistant Professor Universitas Potensi Utama, Medan, Indonesia
2015 – 2019	Lecturer Universitas Potensi Utama, Medan, Indonesia

EDUCATION

- 2021 – 2024 **Doctorate in Mathematics Education**
Universitas Negeri Medan (State University of Medan), Indonesia EQF level 8
- 2013 – 2015 **Master's degree in Mathematics Education**
Universitas Negeri Medan (State University of Medan), Indonesia EQF level 7
- 2012 **Bachelor's degree in Mathematics Education**
Institut Agama Islam Negeri Sumatera Utara (State Islamic Institute of North Sumatra),
Indonesia EQF level 6
-

RESEARCH GRANTS AND FELLOWSHIPS

- Jun 2026 – Nov 2026 **JESH Postdoctoral Fellowship** Principal Researcher
Joint Excellence in Science and Humanities, Austrian Academy of Sciences (OeAW). Host:
Johannes Kepler University Linz. GeoGebra and AI integration in mathematics education;
learning trajectories, research instruments, and Rasch-based analysis of numeracy
learning.
- Aug 2025 – Dec 2027 **RIIM Competition, Wave 7: Neurodiversity-based inclusive mathematics and science
education** Team Member
National Research and Innovation Agency (BRIN), Indonesia. With Universitas
Pendidikan Indonesia and Universitas Tanjungpura. Model development and evaluation
for pre-service teacher preparation; instruments and data analysis.
- Apr 2025 – Dec 2025 **BIMA-DPPM Fundamental Research Grant** Principal Investigator
Kemdiktisaintek, Indonesia. With BRIN, JKU Linz, University of the Philippines, and
Universitas Ahmad Dahlan. Deep learning trajectories and the Ethno-Flipped Classroom
integrated with GeoGebra-AI for numeracy.
- Apr 2024 – Dec 2025 **BIMA-DPPM Domestic Collaborative Research Grant** Principal Investigator
Kemdiktisaintek, Indonesia. Same partners. Differentiated digital learning based on local
culture and the Ethno-Flipped Classroom model for student numeracy.
-

SELECTED JOURNAL ARTICLES

2026

- [1] **Deep learning trajectories for statistical graphs in mathematics learning: Integration of SOLO
taxonomy and ethnomathematics using GeoGebra**
Ramadhani, R., Soeharto, S., Prahmana, R. C. I., Bautista, G., Rahmad, I. F., Rifai, M. Z., Arifiyanti, F., &
Lavicza, Z.
Discover Education.
Q2 · First author · doi:10.1007/s44217-026-01711-7
- [2] **Cultural heritage as a mathematical context: Interpretation of set theory in spatial layout of
Prambanan temple**
With Universitas Ahmad Dahlan and Universitas Sriwijaya.
Infinity Journal.
Q1 · Co-author · Article page

2025

- [3] **Assessing quality and biases in ethnomathematics-based numeracy worksheets: A Many-Facet Rasch
Model analysis**
Ramadhani, R., Soeharto, S., Arifiyanti, F., Prahmana, R. C. I., Saleh, A., & Lavicza, Z.
Social Sciences & Humanities Open, 12, 101736.
Q1 · First author · doi:10.1016/j.ssaho.2025.101736

- [4] **Measuring changes in students' informal statistical reasoning skills through the ethno-flipped classroom model: Stacking and racking analysis**
Ramadhani, R., Soeharto, S., Arifiyanti, F., Lavicza, Z., Syahputra, E., & Simamora, E.
Infinity Journal, 14(4), 949–972.
First author · doi:10.22460/infinity.v14i4.p949-972
- [5] **Enhancing students' informal statistical reasoning skills through the development of a worksheet rooted in the Nias culture**
With Universitas Negeri Medan.
Indexed in ERIC (EJ1474266).
Q2 · First and corresponding author · ERIC record
- [6] **Javanese folklore with moral values: An impactful context in learning relations and functions**
With Universitas Ahmad Dahlan and Universitas Sriwijaya.
Journal on Mathematics Education.
Q1 · Co-author · Article page
- 2024
- [7] **The construct validity of self-regulated learning questionnaire for senior high school students: A Rasch model analysis**
Ramadhani, R., Syahputra, E., & Simamora, E.
International Journal of Evaluation and Research in Education, 13(3), 2032–2038.
Q2 · First and corresponding author · doi:10.11591/ijere.v13i3.26816
- [8] **Integrating traditional food and technology in statistical learning: A learning trajectory**
With Universitas Ahmad Dahlan and Universitas Samudra.
Journal on Mathematics Education, 15(4), 1277–1310.
Q1 · First author · doi:10.22342/jme.v15i4.pp1277-1310
- 2023
- [9] **Merging flipped classroom model with the ethnomathematics approach: A new flexibility learning model**
With Universitas Negeri Medan.
FWU Journal of Social Sciences.
Q1 · First and corresponding author · Article page
- [10] **Expert judgement of collaborative cloud classroom quality and its criteria using the many-facets Rasch model**
With Universitas Negeri Medan and the University of Szeged.
Heliyon.
Q1 · First and corresponding author · Article page
- [11] **Practical and social interaction justice of online learning during the COVID-19 pandemic**
With Universitas Negeri Medan and Universitas Sumatera Utara.
International Journal of Evaluation and Research in Education.
Q2 · First and corresponding author · Article page
- 2022
- [12] **Flipped classroom assisted Autograph in calculus learning for engineering students: A Rasch measurement study**
Ramadhani, R., Bina, N. S., & Syahputra, E.
Mathematics Teaching-Research Journal, 14(4), 36–55.
First author
- [13] **Exploration of students' statistical reasoning ability in the context of ethnomathematics: A study of Rasch model**
Ramadhani, R., Saragih, S., & Napitupulu, E. E.
Mathematics Teaching-Research Journal, 14(1), 138–168.
First author
- Full list on Google Scholar and Scopus.

BOOKS AND CHAPTERS

- 2024 **Students' self-regulated learning in mathematics learning**
Chapter in *Application of Rasch Rating Scale Model*, pp. 6–15. Mind Metric Solution, Selangor, Malaysia.
- 2021 **Statistics in educational research: Mathematical analysis and SPSS applications**
Ramadhani, R., & Bina, N. S. Kencana, Jakarta.

PATENTS AND INTELLECTUAL PROPERTY

- 2025 **GEASTRA: GeoGebra-Supported AI for Statistical Graph Learning** EC002025175169
Copyright, e-book (granted)
- 2025 **eSTEAM-AI: cross-subject STEAM project learning app integrated with AI** EC002025133117
Copyright, computer program (granted)
- 2021 **E-School Management System method** S00202110932
Simple patent (registered)
- 2020 **Hoax-checking application** EC00202029705
Copyright, computer program (granted)
- 2020 **Statistics e-module** EC00202029706
Copyright, e-module (granted)
- 2019 **Smart Check Hoax method** S00202007434
Simple patent (registered)

INVITED TALKS

- 2026 **Focus Group Discussion: OBE curriculum review**
Mathematics Education Study Programme, FKIP UISU
- 2025 **Scientific article writing for Scopus journals**
Teacher Professional Education Study Programme, FKIP Universitas Asahan
- 2024 **Visiting lecturer, local ethnomathematics**
UIN Sumatera Utara
- 2024 **Winning research grants: Tips and tricks from successful lecturers**
OZAMI Inti Sinergi
- 2024 **Doctoral scholarships and developing an academic career**
Dunia Dosen (webinar)
- 2024 **Technical guidance for school principals: Data-based planning and assessment**
Ministry of Education, Culture, Research and Technology, Balai Besar Guru Penggerak North Sumatra

MEMBERSHIPS, SKILLS, AND LANGUAGES

- Memberships **Indonesian Educators Mathematics Society (I-MES), Regional Chair, North Sumatra, 2024–2026.**
Psychology of Mathematics Education (PME), member, 2025–2026.
- Methods **Rasch analysis (WINSTEPS, FACETS, ACER ConQuest 5), structural equation modelling, SAS, Microsoft Excel. Research proposal writing and student assessment.**
- Languages **Bahasa Indonesia (native). English (B2 in listening, reading, speaking, and writing).**